

Containerization : Lab 1

Problem 1

1. Run the container hello-world

```
docker run hello-world
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
\$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
<https://hub.docker.com/>

For more examples and ideas, visit:
<https://docs.docker.com/get-started/>

2. Check container status

```
docker ps -a
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
fb8cdf67cee7	hello-world	"/hello"	About a minute ago	Exited (0) About a minute ago		focused_mcnulty
726c6501a469	github-repos	"python app.py"	8 weeks ago	Exited (137) 7 weeks ago		wizardly_sammet
5b9725486b6c	github-repos	"python app.py"	8 weeks ago	Exited (255) 8 weeks ago	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	practical_euclid
5831f5824d20	github-repos	"python app.py"	2 months ago	Created		wizardly_hamilton
5b4c80ec1d1c	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		naughty_cohen
5f4f0fd4bc1b	github-repos	"python app.py"	2 months ago	Exited (255) 2 months ago	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	unruffled_hofstadter
916b7906c559	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		bold_chaplygin
2af395262797	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		unruffled_heisenberg
9bf448963b34	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		determined_cannon
553bc845d73f	github-repos	"python app.py"	2 months ago	Exited (255) 2 months ago	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	xenodochial_chatelet
246e9c4046f4	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		laughing_hopper
fb864f6bbe1e	github-repos	"python app.py"	2 months ago	Exited (0) 2 months ago		happy_lehmann
2f733122a6ab	github-repos	"python app.py"	2 months ago	Exited (0) 2 months ago		quirky_moser
1c7860acae82	github-repos	"python app.py"	2 months ago	Exited (137) 2 months ago		quizzical_brown

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

3. Start the stopped container

```
docker start fb8cdf67cee7
```

4. Remove the container

```
docker ps -a  
docker rm fb8cdf67cee7
```

5. Remove the image

```
docker ps -a  
docker rmi fb8cdf67cee7
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker start fb8cdf67cee7  
fb8cdf67cee7  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker rm fb8cdf67cee7  
fb8cdf67cee7  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

Problem 2

1. Run container centos or ubuntu in an interactive mode

```
docker run -it ubuntu
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -it ubuntu  
Unable to find image 'ubuntu:latest' locally  
latest: Pulling from library/ubuntu  
01d7766a2e4a: Pull complete  
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9  
Status: Downloaded newer image for ubuntu:latest  
root@25caced926a9:/# exit  
exit  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

2. Run the following command in the container "echo docker "

```
docker run ubuntu echo Hello
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run ubuntu echo Hello
Hello
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

3. Open a bash shell in the container and touch a file named hello-docker

```
docker run -it ubuntu /bin/bash
touch hello.txt
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -it ubuntu /bin/bash
root@2d846c4554c6:/#
root@2d846c4554c6:/# touch hello.txt
root@2d846c4554c6:/#
root@2d846c4554c6:/# exit
exit
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

4. Stop the container and remove it. Write your comment about the file hello-docker

```
docker ps -a
docker stop 2d846c4554c6
docker rm 2d846c4554c6
```

The file will be deleted

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
2d846c4554c6   ubuntu    "/bin/bash"             About a minute ago    Exited (0) 42 seconds ago                               inspiring_gates
3aa795793b6e   ubuntu    "echo Hello"            2 minutes ago        Exited (0) 2 minutes ago                               wonderful_wright
25caced92a69   ubuntu    "/bin/bash"             4 minutes ago        Exited (0) 3 minutes ago                               vibrant_cartwright
726c6501a469   github-repos "python app.py"         8 weeks ago          Exited (137) 7 weeks ago                               wizardly_sammet
5b9725486b6c   github-repos "python app.py"         8 weeks ago          Exited (255) 8 weeks ago                               practical_euclid
5831f5824d20   github-repos "python app.py"         2 months ago         Created                                                wizardly_hamilton
5b4c80ec1d1c   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               naughty_cohen
5f4f0fd4bc1b   github-repos "python app.py"         2 months ago         Exited (255) 2 months ago                               unruffled_hofstadter
916b7906c559   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               bold_chaplygin
2af395262797   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               unruffled_heisenberg
9bf448963b34   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               determined_cannon
553bc845d73f   github-repos "python app.py"         2 months ago         Exited (255) 2 months ago                               xenodochial_chatelet
246e9c4046f4   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               laughing_hopper
fb864f6bb1e1   github-repos "python app.py"         2 months ago         Exited (0) 2 months ago                               happy_lehmann
2f733122a6ab   github-repos "python app.py"         2 months ago         Exited (0) 2 months ago                               quirky_moser
1c7860acae82   github-repos "python app.py"         2 months ago         Exited (137) 2 months ago                               quizzical_brown
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker stop 2d846c4554c6
2d846c4554c6
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker rm 2d846c4554c6
2d846c4554c6
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

5. Remove all stopped containers

```
docker container prune
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] n
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

Problem 3

Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.

```
docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0!
mysql:latest
docker ps
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0! mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
878d1acdc1129ac8463c062f4cabb34e4c29103010ae965728b56673d3dd66d7
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
878d1acdc112   mysql:latest   "docker-entrypoint.s..." 21 seconds ago Up 20 seconds 3306/tcp, 33060/tcp      app-database
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

Problem 4

Run the image Nginx

```
docker run -it --name mynginx nginx
docker ps
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -it --name mynginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
206356c42440: Pull complete
b47f187216b6: Pull complete
1ad233904a11: Pull complete
eedda9fd8786: Pull complete
35ff83c394d6: Pull complete
17d0911leaf62: Pull complete
df0b66c867e4: Pull complete
Digest: sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Status: Downloaded newer image for nginx:latest
```

Add html static files to the container and make sure they are accessible

```
docker exec -it mynginx bash
touch /usr/share/nginx/html/index.html
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
297d49db133b   nginx     "/docker-entrypoint.s..." 5 minutes ago  Up 17 seconds  80/tcp                  mynginx
878dlacdc112   mysql:latest "docker-entrypoint.s..." 8 minutes ago  Up 8 minutes   3306/tcp, 33060/tcp     app-database
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker exec -it mynginx bash
root@297d49db133b:/#
root@297d49db133b:/#
root@297d49db133b:/# touch /usr/share/nginx/html/index.html
root@297d49db133b:/#
root@297d49db133b:/# ls /usr/share/nginx/html/index.html
/usr/share/nginx/html/index.html
root@297d49db133b:/#
```

Commit the container with image name IMAGE_NAME

```
docker commit mynginx
docker images
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker commit mynginx
sha256:3189348a76bd65a1665d3f6f39d9c2815718955370b96304c9ba12764d32504c
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
effect-test:latest	5c0c9f2b5b0c	4.11GB	0B	
elysia-1507-test:latest	8f329132841c	3GB	0B	
ghcr.io/aquasecurity/tracetest:latest	2f779947f973	800MB	0B	
github-repos:latest	a86af5b53634	135MB	0B	U
grafana/grafana:latest	1849e2140421	733MB	0B	
hello-world:latest	1b44b5a3e06a	10.1kB	0B	
hurl-test-enhanced:latest	254aac32c2a4	4.45GB	0B	
hurl-test-final-final:latest	27ef0ab8dd2f	4.45GB	0B	
hurl-test-final:latest	a7e5ed8e4f31	4.45GB	0B	
hurl-test-hardened-v2:latest	b3a083c0147c	4.45GB	0B	
hurl-test-hardened:latest	33b6740a8e2c	4.45GB	0B	
hurl-test-refined-v2:latest	78eefc25f6ea	4.45GB	0B	
hurl-test-refined:latest	8c44b4e52520	4.45GB	0B	
hurl-test:latest	fa662b69b3da	4.45GB	0B	
issue-33895-test:latest	e969c626590b	9.12GB	0B	
maris-4557:latest	4ca5be125f53	4.45GB	0B	
mysql:latest	8458fdcf578c	922MB	0B	U
nginx:latest	fd204fe2f750	161MB	0B	U
process-snapshot-analyzer-process-snapshot-analyzer:latest	9541f87839cd	441MB	0B	
prom/prometheus:latest	4fcecfc061b74	313MB	0B	
ubuntu:latest	bdbabce66f1b	78.1MB	0B	U

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```

Problem 5

Run a container Nginx with name mynginx and attach a volume for containing static html file

```
docker run -d --name mynginx -v myvol:/usr/share/nginx/html
```

Remove the container

```
docker rm -f mynginx
```

Run a new container with the following:

- Attach the volume that was attached to the previous container
- Map port 80 to port 9898 on you host machine
- Access the html files from your browser

```
docker run -d --name mynginx -v myvol:/usr/share/nginx/html -p 9898:80  
nginx  
curl localhost:9898
```

```
mynginx  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -d --name mynginx -v myvol:/usr/share/nginx/html -p 9898:80 nginx  
e4b44769eac047836c02bb507a03f290365f523f37faa1f0093efea0299a693c  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker rm -f mynginx  
mynginx  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ docker run -d --name mynginx -v myvol:/usr/share/nginx/html -p 9898:80 nginx  
f0ed1635303ef95e3d5d7fe52aa77d89c6cf24a7ceb9e16989a280c2b2f87e5d  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$ curl localhost:9898  
<!DOCTYPE html>  
<html>  
<head>  
<title>Welcome to nginx!</title>  
<style>  
html { color-scheme: light dark; }  
body { width: 35em; margin: 0 auto;  
font-family: Tahoma, Verdana, Arial, sans-serif; }  
</style>  
</head>  
<body>  
<h1>Welcome to nginx!</h1>  
<p>If you see this page, the nginx web server is successfully installed and  
working. Further configuration is required.</p>  
  
<p>For online documentation and support please refer to  
<a href="http://nginx.org/">nginx.org</a>.<br/>  
Commercial support is available at  
<a href="http://nginx.com/">nginx.com</a>.</p>  
  
<p><em>Thank you for using nginx.</em></p>  
</body>  
</html>  
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions$
```